

Inclusive Quiet Room -for building an inclusive society-

Shoko Kimura
kshoko@siggraph.org
The University of Tokyo / Nagoya
Institute of Technology
Tokyo, Japan

Kenichi Ito
itokenichi@lelab.t.u-tokyo.ac.jp
The University of Tokyo
Chiba, Japan

Ayaka Fujii
a-fujii@aist.go.jp
National Institute of Advanced
Industrial Science and Technology
Tokyo, Japan

Rihito Tsuboi
r.tsuboi.contact@gmail.com
Game Sound Creator
Tokyo, Japan

Kazuki Okawa
k_okawa@vrlab.esys.tsukuba.ac.jp
University of Tsukuba
Tsukuba, Japan

Hibiki Kojima
hibiki9358810@gmail.com
Musik und Kunst Privatuniversität
der Stadt Wien
Vienna, Austria

Keisuke Kitagawa
keisuke@eelll.jp
Nagoya Institute of Technology
Nagoya, Japan

Yoshinori Natsume
yn@nitech.ac.jp
Nagoya Institute of Technology
Nagoya, Japan

ABSTRACT

A large percentage of people with autism or developmental disorders, which are mental disabilities, have sensory hypersensitivity. Therefore, the spread of “quiet rooms” in which they can feel at ease in social life is a necessary element in realizing a symbiotic society. However, the high cost of installing quiet rooms, which require highly soundproof rooms isolated from the outside, is an obstacle to their widespread use. The Inclusive Quiet Room is a new concept of portable quiet rooms that combines an easy-to-construct instant house, immersive videos, and relaxing sounds. In addition to enabling many people to experience the benefits of the room, the work proposes an image of the future quiet rooms that can be easily constructed anywhere. In this paper, we analyze the effectiveness of the Inclusive Quiet Room, exhibited in France, based on survey data from 372 respondents. Through the analysis, the relaxation effects and the demands for quiet rooms are substantiated. The room gives the feeling of being warmly embraced and secured. If all people including those without mental disorders could experience this embraced feeling, they would understand the need and benefits of relaxing environments for the people with sensory hypersensitivities.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility technologies**; *Virtual reality*.

KEYWORDS

Autism and developmental disabilities, Environmental adjustment, Quiet Room, Sensory sensitivity, Snoezelen, Sensory stimulation

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Figure 1: Inclusive Quiet Room. Left: The University of Tokyo iii Exhibition. Right: Laval Virtual 2023.

1 INTRODUCTION

The “quiet room” is a space to offer a calming environment primarily for individuals with sensory hypersensitivity and mental disabilities, such as autism and developmental disorders. The purpose of this room is to reduce external stimuli, thus preventing panic attacks induced by sensory hypersensitivity [Karita 2009]. The room can also be referred to as a sensory room [Haig and Hallett 2023; Unwin et al. 2022], or a darkroom, and sometimes the notation for it is lengthy. In this paper, we use the term “quiet room” for the sake of brevity and clarity.

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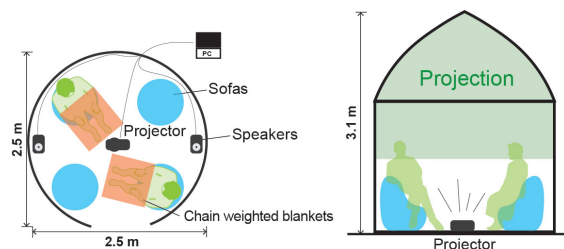


Figure 2: Arrangement of Inclusive Quiet Room.

2 METHOD

2.1 Overview of the Concept

We created the Inclusive Quiet Room by combining the principles of traditional quiet rooms with digital rehabilitation, incorporating tactile, auditory, and visual stimuli (Figure 1, 2). This installation was designed based on the opinions of individuals with autism and developmental disabilities. The purpose of this experience is to bring relaxation and help people regain their calm. The Inclusive Quiet Room can be easily implemented by using a simple building called instant house that is a space soundproofed by insulation, allowing for casual experiences in settings such as exhibition venues. It is possible to convey the necessity and benefits of the quiet rooms to all people regardless of whether they have sensory hypersensitivity or not, fostering a broader societal understanding and acceptance. This simplicity and accessibility also provide a potential for effective societal implementation of quiet rooms in the future.

2.2 User Experience

Participants enter the Inclusive Quiet Room, where they can sit on soft cushions and, if they wish, cover themselves with the chain-weighted blanket. They can then immerse themselves in an experience that combines a projected video of starry skies and psychedelic effects with a musical blend of piano and natural sounds. Although the video and music content lasts approximately 10 minutes, participants are free to enter and exit the installation at their discretion. After the experience, participants are asked to fill out a questionnaire regarding their impressions and feedback.

3 QUESTIONNAIRE RESULTS

We report on the questionnaire results obtained during Laval Virtual, an exhibition of Virtual Reality (VR) held in France, spanning from April 12th to April 16th, 2023 (Figure 1 Right). The survey collected a total of 372 responses, consisting of 177 males, 184 females, and 11 others. Figure 3 shows a portion of the survey results. Each question was evaluated using a 7-point Likert scale, ranging from 1 (not at all applicable) to 7 (highly applicable). In the item "I relaxed after watching the psychedelic video," approximately 68% of respondents provided a positive response of 5 or higher. In the item "I was comforted by the embrace of the weighted blanket," approximately 81% provided a positive response. In terms of the overall experience, 90% of respondents provided a positive response to the question "I was able to calm down and relax," and 79% thought positively about the statement "Society needs more quiet rooms." The results have demonstrated that our work has the potential to provide a healing

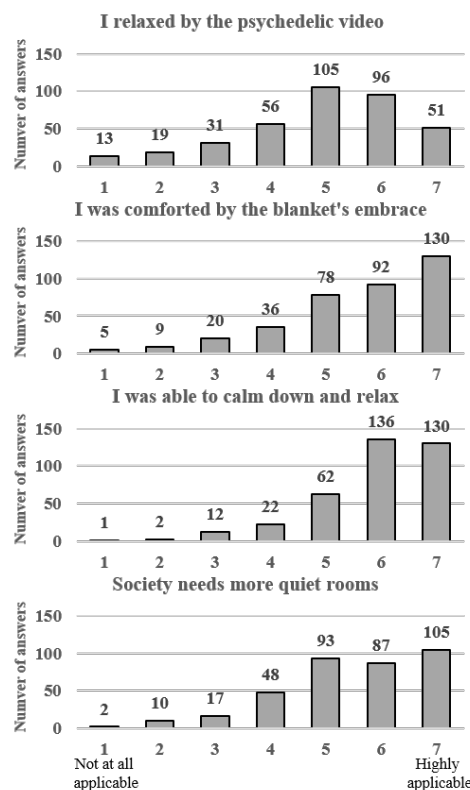


Figure 3: Questionnaire results.

effect to many people. Furthermore, the results have affirmed the importance of implementing quiet rooms in society.

4 CONCLUSION

We proposed the Inclusive Quiet Room, which is a portable quiet room designed for people who suffer from sensory hypersensitivity and panic attacks to take a break and relax. It combines an easy-to-construct instant house, immersive videos, relaxing sounds, soft cushions, and chain-weighted blankets. Through the survey, we confirmed the effectiveness and the demand of the proposed method. In the future, it is crucial to increase the number of places where people can relax in a society to reduce the occurrence of panic.

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